Endogenous Credibility in Social Learning

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Abstract

Connected groups sometimes learn faster than isolated individuals, yet sometimes converge confidently on the wrong answer. We argue that part of the explanation is that social signals are not neutral: they are weighted by a *credibility* that is generated endogenously by the decision process itself. We develop an agent-based model in which each agent is a reinforcement learner whose binary choices arise from a drift–diffusion process. The same within-trial process produces an endogenous confidence signal, and confidence acts as a credibility weight in two distinct social channels: an *anticipatory* channel, through which neighbours' past confidence-weighted actions enter subjective value before choice, and a *retrospective* channel, through which neighbours' observed outcomes modulate value revision after choice. Agents are embedded in community-structured networks; under balanced block weights the anticipatory social field admits an exact quotient representation over communities (for any number of communities). We implement the model, verify it against a ten-point test contract, and study it by Monte Carlo simulation in two-community societies and a four-community modular extension. The simulations identify four interpretable regimes — efficient consensus, wrong consensus, polarisation and unresolved learning — and show that connectivity is *non-monotone*: moderate credibility-weighted transmission corrects error and accelerates consensus, whereas strong transmission amplifies early overconfident mistakes. Whether the distortion takes the form of society-wide wrong consensus or of persistent between-community divergence is governed by cross-community permeability. Decisive ablations, identified by common random numbers, show that the wrong-consensus regime is not a generic consequence of social learning: at the contested operating point, removing the anticipatory channel, the retrospective channel, or the decision-time dependence of confidence eliminates it; removing the credibility weighting weakens but does not eliminate it; and constant learning rates intensify it. We report Wilson and bootstrap intervals throughout and confirm the phase structure against threshold, horizon, reward-gap, initial-condition, grid-resolution and random-graph perturbations. The contribution is to show that connectivity is conditionally beneficial, and that its sign depends on whether credibility tracks accuracy or amplifies early error. A complete replication package reproduces every figure and table.

Keywords: social learning; credibility; confidence; drift–diffusion model; reinforcement learning; community networks; agent-based simulation; quotient aggregation; Monte Carlo.

1 Introduction

A long tradition in the social and economic sciences holds that connecting learners is broadly beneficial: pooling dispersed information should, on average, move groups towards better collective beliefs (DeGroot, 1974; Golub and Jackson, 2010). Yet the same literature documents the opposite outcome just as often. Informational cascades can lock groups onto incorrect actions (Bikhchandani et al., 1992), crowds can be wise or mad depending on subtle features of how information is shared (Tump et al., 2020), and reward-like social feedback can entrench polarisation even on fixed networks (Horn et al., 2026). The unresolved question is therefore not *whether* connectivity helps, but *when*: under what conditions does adding social links correct individual error, and under what conditions does it amplify it into confident collective mistakes or persistent divergence between groups? Our central hypothesis is that the answer is decided by an endogenous, socially transmitted variable — the *credibility* of a signal, generated by the sender's own decision confidence — so that connectivity corrects error when credibility tracks accuracy and amplifies it when credibility tracks early, possibly mistaken, conviction (Figure 1 sketches the mechanism).

We propose that an important part of the answer lies in what is actually transmitted. Agents rarely exchange bare actions or unweighted observations. They transmit actions and outcomes that arrive marked by a sense of how *credible* they are, and that credibility is not exogenous: it is generated by the very process that produced the action. We formalise this as *endogenous credibility*. Confidence is produced by an agent's own decision process; once it is socially interpreted as credibility, it weights how strongly a neighbour's behaviour influences others. Because confidence can attach to correct *or* to incorrect early choices, the same network links can transmit correction or amplify error. Connectivity is then no longer uniformly beneficial: its effect depends on whether the circulating credibility tracks accuracy or tracks early, possibly mistaken, conviction. Its sign is set by the credibility-generating process behind transmitted behaviour, not by connection itself.

The model makes this precise with deliberately minimal behavioural machinery. Each agent carries latent action values that are updated from reward prediction errors (Sutton and Barto, 2018; Dayan and Daw, 2008). Choices are not produced by a static policy but by a drift–diffusion process, so that action, decision time and confidence are jointly determined by the strength of accumulated evidence (Ratcliff, 1978; Ratcliff et al., 2016). Confidence is read out from that same process (Kiani and Shadlen, 2009; Drugowitsch, 2019). Social influence operates through two channels that we keep strictly separate: an anticipatory channel that enters subjective value before the decision, and a retrospective channel that enters value revision after the outcome is observed. Agents live in modular networks whose community structure is summarised by a row-stochastic coupling matrix.

This is an agent-based computational social simulation in the sense emphasised by recent work in this journal: interactions and aggregates emerge from explicit behavioural rules rather than being imposed as exogenous summaries (Stone et al., 2026), the design is documented following the ODD protocol and its decision-oriented extension (Grimm et al., 2010, 2020; M"uller et al., 2013), and the baseline is kept narrow so that its parameters remain identifiable and its observables remain informative (Olukan et al., 2026; Carrella, 2021; McCulloch et al., 2022). The model is implemented, verified against an explicit test contract, and studied by Monte Carlo simulation; a replication package regenerates every result.

Contributions. The novelty is not the ingredients — reinforcement learning, drift–diffusion choice, confidence, and social learning are each established — but their coupling: confidence is generated by the *same* decision process that produces the action, it regulates *both* private and social learning, and credibility is not a fixed edge weight but an *emergent* property that is produced by neighbours' decisions and propagates through the network. Concretely, the paper makes five contributions.

(i) It introduces an agent-based model in which confidence is generated by a process-based choice rule rather than assigned exogenously, and is transmitted socially as credibility.

(ii) It distinguishes anticipatory social influence from retrospective social learning and keeps the two channels separate in prose, equations, code and ablations.

(iii) It embeds agents in modular networks and proves that, under balanced block weights, the anticipatory social field admits an *exact* quotient representation over communities.

(iv) Using simulation, it identifies when connectivity corrects error, when it amplifies early overconfident mistakes into wrong consensus, and when it produces between-community polarisation, mapping these regimes in a connectivity phase diagram.

(v) It validates the community-level quotient description against the full agent-level model and delimits where the reduction is faithful.

2 Background and positioning

Social learning, herding and networked learning. The primary anchor is the literature on collective learning and opinion formation in networks (Golub and Sadler, 2016). Classical models of consensus and naive learning explain when decentralised updating aggregates information efficiently (DeGroot, 1974; Friedkin and Johnsen, 1990; Golub and Jackson, 2010), and continuous-opinion and bounded-confidence models explain when interaction yields consensus, clustering or persistent disagreement (Deffuant et al., 2000; Hegselmann and Krause, 2002; Flache et al., 2017). The herding and cascade literature explains how rational or boundedly rational agents can lock onto incorrect actions when they over-weight others' behaviour relative to their own signal (Banerjee, 1992; Bikhchandani et al., 1992; Tump et al., 2020). Cooperative and networked multi-armed bandit models study the same tension between social acceleration and distortion in an explicit reward-learning setting, including on block-structured graphs (Landgren et al., 2021; Xu et al., 2025), and reward-like social feedback can drive polarisation directly even on fixed networks (Banisch and Olbrich, 2019; Horn et al., 2026). Our model contributes a mechanism to this tradition: the weight an agent places on a neighbour is not fixed but is an endogenous credibility produced by that neighbour's own decision process.

What the model changes relative to adjacent traditions. Table 1 positions the model. Relative to DeGroot and bounded-confidence opinion models, agents do not update opinions by weighted averaging alone: they learn from stochastic rewards, choose through a process model, and transmit confidence-marked actions and outcomes. Relative to herding and cascade models, the credibility attached to social information is not exogenously given but generated by the neighbour's own decision process. Relative to cooperative bandit models, the model separates anticipatory influence on choice from retrospective influence on value revision. Relative to polarisation ABMs, modular divergence arises from the interaction between confidence-weighted transmission and reward learning rather than from fixed homophily or ideological repulsion alone; and relative to reinforcement-learning drift–diffusion (RLDDM) accounts of individual choice, the novelty is the social transmission of endogenous credibility on a network.

Agent-based social simulation and documentation. Methodologically the paper follows current practice in computational social simulation. A recurring theme is that the structures of interest should emerge from explicit mechanisms rather than be imposed as aggregates (Stone et al., 2026), and that model purpose, design concepts and implementation should be documented separately (Grimm et al., 2010, 2020; M"uller et al., 2013). A second theme is discipline about heterogeneity, calibration and observables: descriptive richness can degrade identifiability if introduced faster than the inferential apparatus that supports it (Olukan et al., 2026; Windrum et al., 2007; Carrella, 2021; McCulloch et al., 2022; Polhill et al., 2021). We therefore begin with a homogeneous two-community baseline and add structure only after the mechanism is isolated.

Table 1: Positioning relative to adjacent model families.

Tradition	Typical mechanism	Exogenous element	What this paper changes
DeGroot / consensus	weighted opinion averaging	weights, signals	confidence generated by the choice process
Herding / cascades	action inference	reliability of social signal	credibility is endogenous
Cooperative bandits	reward learning across agents	communication weight	two distinct social channels
Polarisation ABMs	homophily / social feedback	confidence usually absent	confidence-driven amplification
RLDDM (individual)	value-based diffusion choice	no social network	social transmission of credibility

Update scheduling is treated as a modelling decision rather than an implementation detail, since activation regimes can alter dynamics (Axtell, 2000; Caron-Lormier et al., 2008; Weimer et al., 2019); we fix synchronous trial timing with a simultaneous post-outcome update. We also represent modular structure through interpretable community-coupling parameters rather than an arbitrary graph family, in the spirit of stratified and block-structured network models (Gonçalves et al., 2026; Lee and Wilkinson, 2019; Peixoto, 2014).

Computational models of social cognition and reinforcement learning. Contemporary computational accounts treat social learning with the same primitives as non-social value learning — value, prediction error, uncertainty and learning rate (Ruff and Fehr, 2014; Lockwood and Klein-Fl"ugge, 2021). This licenses our decision to let social information enter the same latent-value machinery as private reward, and to allow learning rates that depend on confidence rather than being globally fixed (Wang et al., 2023). In this reading the retrospective channel is an *other-referenced* prediction error, the social counterpart of the self-referenced reward prediction error, consistent with evidence that social and non-social learning share prediction-error machinery in the brain (Joiner et al., 2017).

Process-based choice and confidence. Drift–diffusion and reinforcement-learning-diffusion models make choice a within-trial accumulation process and render decision time informative (Ratcliff, 1978; Ratcliff et al., 2016; Pedersen et al., 2017; Fontanesi et al., 2019). The diffusion limit is statistically optimal for two-alternative choice (Bogacz et al., 2006) and admits an economic optimal-stopping foundation in which the decision time is itself chosen (Fudenberg et al., 2018). Confidence can be read from the same accumulation process (Kiani and Shadlen, 2009; Pleskac and Busemeyer, 2010; Drugowitsch, 2019), can regulate learning, and may remain miscalibrated (Salem-Garcia et al., 2023); communicated confidence can even improve joint decisions (Bahrami et al., 2010), which is exactly the channel that the present model makes double-edged. We use these results as modelling restrictions: confidence is endogenous, depends on evidence strength and decision time, regulates learning, and is *not* equated with accuracy.

Why confidence is socially transmitted. Three questions motivate the mechanism. *Why should confidence be transmitted socially?* Because confidence is a compact, decision-general summary of evidence reliability that a receiver can act on without re-deriving the sender's computation — the role of a communicable teaching signal. *Which computations are shared between brains?* Social and non-social learning recruit common value and prediction-error

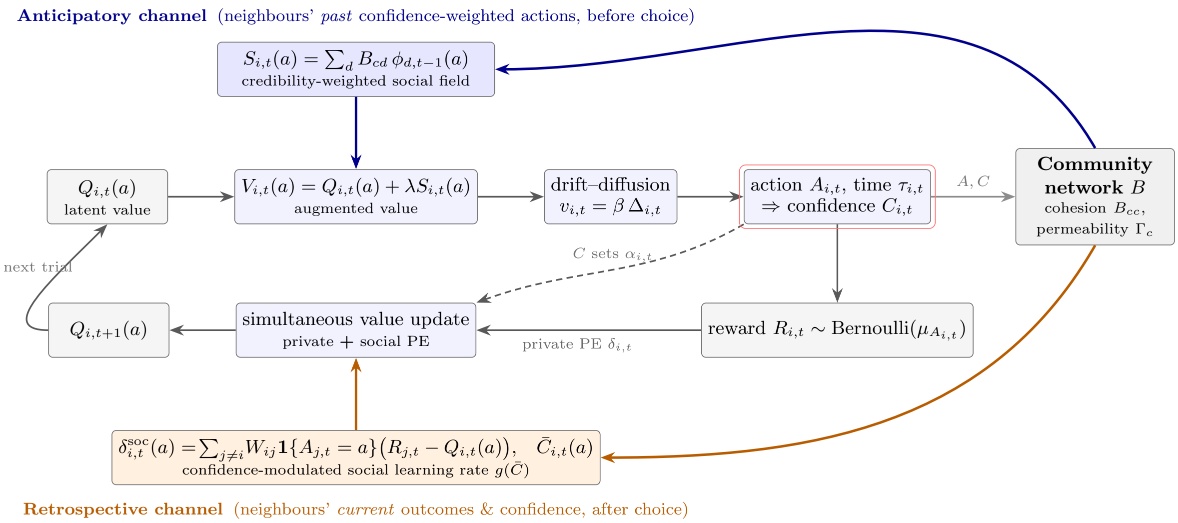

Figure 1: Mechanism. Within a trial, latent values $Q_{i,t}$ are augmented by the anticipatory credibility field $S_{i,t}$ (top, blue) to form subjective values $V_{i,t}$; a drift–diffusion process turns the value contrast into an action $A_{i,t}$, a decision time $\tau_{i,t}$ and an endogenous confidence $C_{i,t}$ (red box). After the reward is observed, values are revised by a simultaneous update combining the private prediction error with the retrospective social prediction error (bottom, orange), whose strength is modulated by neighbours' confidence. Confidence is the shared link between the two channels. The community coupling matrix B governs how credibility circulates.

machinery, so an other-referenced prediction error is the natural social analogue of the reward prediction error (Joiner et al., 2017; Lockwood and Klein-Fl"ugge, 2021). *How does this relate to teaching signals?* Orbitofrontal adaptive-learning accounts emphasise value computation, task-state representation, uncertainty-sensitive updating and teaching-like feedback (Schuck et al., 2016; Behrens et al., 2018; Niv, 2019; Wang et al., 2023; Rao et al., 2025); our confidence-modulated learning rule is such a teaching-like signal, routed *between* agents rather than within one brain. The literature motivates the mechanism; the claims here are computational and behavioural.

3 The model

We describe the model following the logic of the ODD protocol; the full ODD+D description is given in Appendix B. Figure 1 summarises the within-trial mechanism and the two social channels.

3.1 Purpose and entities

The model's purpose is to explain when credibility-weighted social transmission makes connectivity corrective and when it makes connectivity distortive. The entities are *agents* and *communities*. Agents choose repeatedly between two arms of a stochastic bandit, learn from reward, and influence one another through a weighted communication network. Communities are an exogenous partition of agents into modules.

3.2 State variables and environment

Time is discrete at the trial level, $t \in \mathbb{N}_0$. Agents are $V = \{1, \ldots, N\}$, partitioned into communities $V = \bigsqcup_{c=1}^{M} V_c$ with $N_c = |V_c|$; community membership is $g(i)$. The action set is $\mathcal{A} = \{1, 2\}$. The weighted communication matrix $W = (W_{ij}) \in \mathbb{R}_+^{N\times N}$ is row-stochastic, $\sum_j W_{ij} = 1$. For each agent i, trial t and arm a we track the latent value $Q_{i,t}(a)$, the anticipatory social signal $S_{i,t}(a)$, the augmented value $V_{i,t}(a)$, the value contrast $\Delta_{i,t}$, the drift $v_{i,t}$, the action $A_{i,t}$, the reward $R_{i,t} \in \{0, 1\}$, the decision time $\tau_{i,t}$ and the confidence $C_{i,t} \in (0, 1)$. Arm a pays Bernoulli reward with mean $\mu_a \in (0, 1)$; $a^\star \in \arg\max_a \mu_a$, $\mu^\star = \max_a \mu_a$ and $\Delta\mu = \mu_1 - \mu_2$.

3.3 Process overview and scheduling

Each trial uses a synchronous schedule with a simultaneous post-outcome update: (1) construct the anticipatory social signal from the previous period; (2) form augmented values; (3) generate the drift–diffusion choice and decision time; (4) read out confidence; (5) draw rewards; (6) compute all private and social prediction errors from the *pre-update* values; (7) apply private and social updates simultaneously. Step 6 prevents order artefacts between private and social revision.

Information structure. Before choosing, agents observe neighbours' previous actions and confidence signals. After outcomes are realised, they observe neighbours' actions, realised outcomes and confidence for the retrospective channel. Confidence is interpreted as communicated or behaviourally inferred subjective reliability, not as objective accuracy.

3.4 Submodels

Anticipatory social signal. Social information available before the current choice is

$$S_{i,t}(a) := \sum_{j=1}^{N} W_{ij}\, \mathbf{1}\{A_{j,t-1} = a\}\, C_{j,t-1}, \qquad S_{i,0}(a) = 0, \tag{1}$$

the confidence-weighted mass of neighbours who chose a in the previous trial. Subjective values are socially augmented,

$$V_{i,t}(a) := Q_{i,t}(a) + \lambda\, S_{i,t}(a), \qquad \Delta_{i,t} := V_{i,t}(1) - V_{i,t}(2), \tag{2}$$

where $\lambda \geq 0$ is the anticipatory weight.

Process-based choice and decision time. Conditional on $\Delta_{i,t}$, the within-trial decision variable is a Wiener process with drift,

$$\mathrm{d}X_{i,t}(s) = v_{i,t}\, \mathrm{d}s + \sigma\, \mathrm{d}B_{i,t}(s), \quad X_{i,t}(0) = 0, \quad v_{i,t} = \beta\, \Delta_{i,t}, \tag{3}$$

absorbed at symmetric boundaries $\pm a$; the action is $A_{i,t} = 1$ if the upper boundary is hit first and $A_{i,t} = 2$ otherwise, with first-passage time $\tau_{i,t}$. We document the implementation explicitly because confidence depends on both drift and decision time. The choice probability is computed *exactly* from the Wiener first-passage solution,

$$\mathbb{P}(A_{i,t} = 1) = \frac{1}{1 + \exp(-2 v_{i,t} a / \sigma^2)}, \tag{4}$$

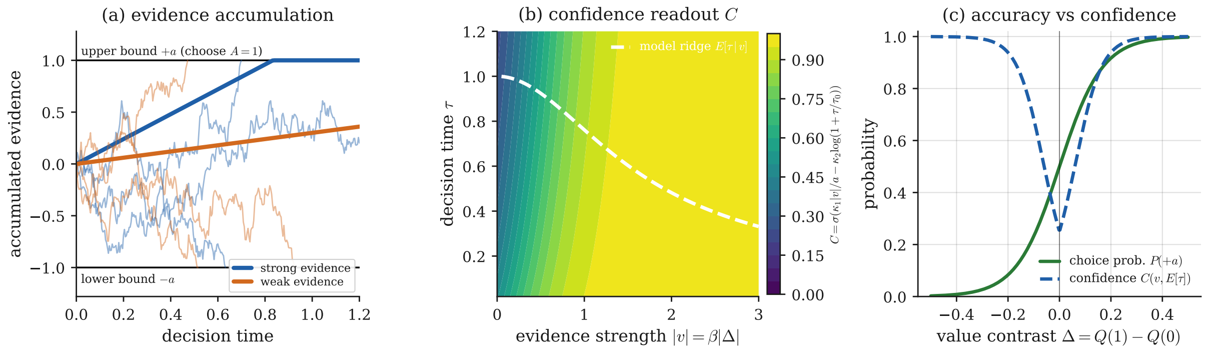

Figure 2: Endogenous confidence as a decision-process readout (all curves are the exact model maps, not fits). **(a)** Evidence accumulates to a boundary $\pm a$ at rate $v=\beta\Delta$; strong evidence yields fast, confident choices, weak evidence slow, uncertain ones. **(b)** Confidence $C=\Lambda(\kappa_1|v|/a-\kappa_2\log(1+\tau/\tau_0))$ rises with evidence strength and falls with decision time; the dashed ridge is the process's own mean decision time $\mathbb{E}[\tau\mid v]$. **(c)** Choice accuracy $P(+a)$ is monotone in the value contrast, but confidence is *U-shaped* in Δ — it tracks evidence *magnitude*, not correctness — so a strongly held wrong belief ($\Delta<0$, large) is transmitted with high credibility. This decoupling of confidence from accuracy is the neuroeconomic root of high-confidence error.

The choice rule (4) and the mean first-passage time are *exact* (Proposition 1). The joint choice–time law, however, is an *approximation*: reaction times are drawn from a positive, mean-matched inverse-Gaussian whose mean equals $\mathbb{E}[\tau_{i,t}]$ and whose squared coefficient of variation is a fixed dispersion, rather than from the exact two-boundary first-passage density. Appendix C validates the choice probability and the mean reaction time against a direct Euler–Maruyama simulation of (3), and Appendix F shows that regime classification is insensitive to this approximation.

Proposition 1 (Exact choice probability and mean decision time)**.** *For the diffusion* (3) *with constant drift $v\neq 0$, infinitesimal variance σ^2 and symmetric absorbing boundaries $\pm a$ started at the origin, the probability of absorption at $+a$ and the expected absorption time are*

$$\mathbb{P}(A=1)=\frac{1}{1+e^{-2va/\sigma^2}},\qquad \mathbb{E}[\tau]=\frac{a}{v}\tanh\Big(\frac{av}{\sigma^2}\Big),$$

with $\mathbb{E}[\tau]\to a^2/\sigma^2$ as $v\to 0$.

Proofs of Propositions 1, 2 and 3 are collected in Appendix A.

Endogenous confidence. Confidence is read out from the same process and is bounded in $(0,1)$:

$$C_{i,t}=\Lambda\bigg(\kappa_1\frac{|v_{i,t}|}{a}-\kappa_2\log\Big(1+\frac{\tau_{i,t}}{\tau_0}\Big)\bigg),\qquad \Lambda(x)=\frac{1}{1+e^{-x}},\tag{5}$$

with $\kappa_1,\kappa_2,\tau_0>0$. Confidence rises with evidence strength and falls with decision time. It is a subjective reliability weight and is *not* assumed to equal correctness; this separation is what allows high-confidence error. An alternative balance-of-evidence map, in which confidence equals the DDM probability of the chosen boundary, is used in the ablations.

Private value revision. The private prediction error is $\delta_{i,t}=R_{i,t}-Q_{i,t}(A_{i,t})$, with a confidence-modulated learning rate

$$\alpha_{i,t}=\begin{cases}\alpha^{\min}+(\alpha^{\max}-\alpha^{\min})\,C_{i,t}, & \delta_{i,t}<0,\\ \alpha^{\max}-(\alpha^{\max}-\alpha^{\min})\,C_{i,t}, & \delta_{i,t}\geq 0,\end{cases}\tag{6}$$

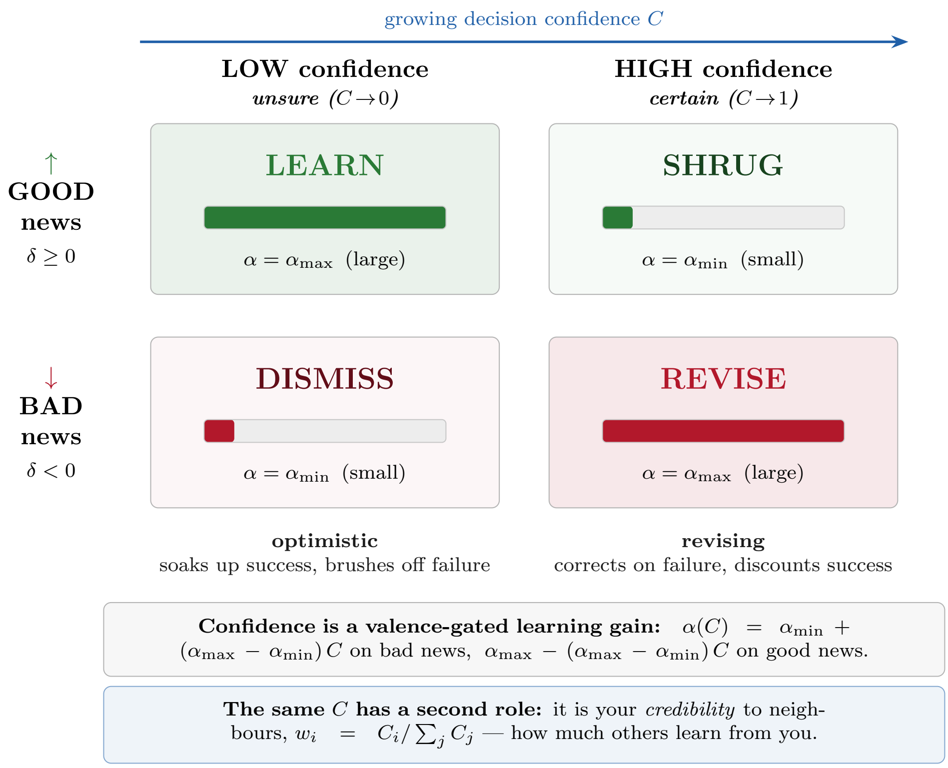

Figure 3: Confidence is a valence-gated learning gain. Reading the 2×2 matrix: the confidence-modulated learning rate (6) is largest ($\alpha_{\max}$) for *good* news under *low* confidence and for *bad* news under *high* confidence, and smallest ($\alpha_{\min}$) on the other diagonal. A low-confidence agent is therefore optimistic (soaks up success, brushes off failure); a high-confidence agent is revising (corrects on failure, discounts success) — a confidence-gated analogue of asymmetric reward-prediction-error learning. The same confidence C plays a second role in social transmission: it is the sender's credibility weight $w_i = C_i / \sum_j C_j$, so confidence is simultaneously an intra-agent gain and an inter-agent edge weight. The exact curves are plotted in Appendix E (Figure 14).

so confident mistakes induce larger corrections and confident successes induce smaller revisions, $0 < \alpha^{\min} \leq \alpha^{\max} < 1$.

Retrospective social learning. The retrospective channel uses neighbours' realised outcomes. For each arm,

$$\delta^{\mathrm{soc}}_{i,t}(a) := \sum_{j \neq i} W_{ij}\, \mathbf{1}\{A_{j,t} = a\}\big(R_{j,t} - Q_{i,t}(a)\big), \qquad \bar{C}_{i,t}(a) := \frac{\sum_{j \neq i} W_{ij}\, \mathbf{1}\{A_{j,t} = a\}\, C_{j,t}}{\sum_{j \neq i} W_{ij}\, \mathbf{1}\{A_{j,t} = a\} + \varepsilon}. \tag{7}$$

We adopt the convention that *confidence modulates the social learning rate* (rather than re-weighting the social prediction error itself): the social learning rate is $g(\bar{C}_{i,t}(a)) = \gamma\, \bar{C}_{i,t}(a)^{\omega}$ with $\gamma > 0, \omega > 0$, $g(0) = 0$. The simultaneous law of motion, with all prediction errors formed from $Q_{i,t}$, is

$$Q_{i,t+1}(a) = \Pi_{[0,1]}\Big(Q_{i,t}(a) + \mathbf{1}\{A_{i,t} = a\}\, \alpha_{i,t}\, \delta_{i,t} + \eta\, g(\bar{C}_{i,t}(a))\, \delta^{\mathrm{soc}}_{i,t}(a)\Big), \tag{8}$$

where $\eta \geq 0$ is the retrospective weight and $\Pi_{[0,1]}$ projects onto the reward support, keeping values interpretable as Bernoulli-mean estimates.

Self-weighting convention. We exclude an agent's own current outcome from the retrospective channel (the sum in (7) is over $j \neq i$), since that outcome is already captured by the private prediction error. We *include* the agent's own previous confidence-weighted action in the anticipatory signal (1) (so $W_{ii} \geq 0$), interpreting it as choice inertia. This convention makes the quotient field of Section 4 exact; the verification suite tests both inclusions (Appendix C).

4 Two communities and the quotient field

We first study the smallest non-trivial modular society and then generalise. For agent $i \in V_c$ and community d, the block exposure is $b_{i\to d} := \sum_{j\in V_d} W_{ij}$, with $\sum_d b_{i\to d} = 1$.

Assumption 1 (Balanced block weights)**.** *There is a row-stochastic matrix $B = (B_{cd})_{c,d=1}^{M}$ with $W_{ij} = B_{cd}/N_d$ for all $i \in V_c$, $j \in V_d$.*

The central mesoscopic object is the confidence-weighted action mass

$$\phi_{c,t}(a) := \frac{1}{N_c} \sum_{i\in V_c} \mathbf{1}\{A_{i,t} = a\}\, C_{i,t}, \tag{9}$$

which combines the fraction choosing a with the average confidence carried by those choices; we also record the action frequency $m_{c,t}(a) = \frac{1}{N_c} \sum_{i\in V_c} \mathbf{1}\{A_{i,t} = a\}$ and the mean value $\bar{Q}_{c,t}(a)$.

Proposition 2 (Exact quotient social field)**.** *Under Assumption 1, for every $i \in V_c$, arm a and $t \geq 1$,*

$$S_{i,t}(a) = \sum_{d=1}^{M} B_{cd}\, \phi_{d,t-1}(a). \tag{10}$$

In particular the anticipatory social field is identical for all agents in a community and is a function only of the community-level masses $\phi_{d,t-1}$.

Equation (10) is the precise sense in which the community network is a quotient of the microscopic network for the anticipatory channel. The matrix B is the community coupling matrix: its diagonal B_{cc} is *cohesion*, $\Gamma_c := 1 - B_{cc} = \sum_{d\neq c} B_{cd}$ is cross-community *permeability*, and $\kappa(B) := 1 - |\lambda_2(B)|$ is the mesoscopic mixing rate. Computationally, (10) also makes the agent-level simulation cheap: under balanced weights both social channels reduce to community aggregates with an $O(1/N_c)$ self-correction, so the exact dynamics run in $O(N)$ rather than $O(N^2)$ per trial.

Defining the community contrasts $\bar{\Delta}^Q_{c,t} = \bar{Q}_{c,t}(1) - \bar{Q}_{c,t}(2)$ and $\bar{\Delta}^S_{c,t} = \sum_d B_{cd}(\phi_{d,t-1}(1) - \phi_{d,t-1}(2))$, the total mesoscopic contrast $\bar{\Delta}_{c,t} = \bar{\Delta}^Q_{c,t} + \lambda\bar{\Delta}^S_{c,t}$ exposes the model's central tension: the private component is disciplined by realised reward differences, while the social component amplifies whatever confidence-weighted mass is already present.

Remark 1 (Exact field, approximate value dynamics)**.** *The field (10) is exact under Assumption 1. The value dynamics, however, remain stochastic at the agent level because rewards and diffusion noise are individual. A community-level (representative-agent) description is therefore exact for the anticipatory field but only approximate for the full value process. We quantify this approximation in Section 6.6.*

The quotient also yields an analytical handle on the model's central non-monotonicity. Linearising the deterministic community recursion around the symmetric indifference state isolates the anticipatory feedback loop.

Proposition 3 (Local amplification threshold for the anticipatory loop)**.** *Linearise the deterministic community recursion about the symmetric indifference state ($\bar{\Delta}^Q_{c,t} = 0$, equal masses, equal arms). Let $\kappa = \beta a/\sigma^2$ be the choice sensitivity, C_0 the confidence at the indifference point, and $\bar{b}$ the effective within-community coupling. Then the confidence-weighted mass gap $\psi_{c,t} = \phi_{c,t}(1) - \phi_{c,t}(2)$ obeys, to first order,*

$$\psi_{c,t} = C_0 \kappa\, \bar{\Delta}_{c,t}, \qquad \bar{\Delta}_{c,t} = \bar{\Delta}^Q_{c,t} + \lambda\, \bar{b}\, \psi_{c,t-1},$$

so the anticipatory loop has one-step multiplier $\rho(\lambda) = C_0\, \kappa\, \bar{b}\, \lambda$. The loop is contractive iff $\lambda < \lambda^\star := \sigma^2/(\beta a\, C_0\, \bar{b})$. For $\lambda < \lambda^\star$ the social field self-limits and tracks the reward-disciplined value gap (connectivity corrective); for $\lambda > \lambda^\star$ an early mass gap is amplified geometrically faster than private learning can remove it (connectivity distortive).

For the default parameters ($\beta = 6$, $\sigma = a = 1$, indifference confidence $C_0 \approx 0.25$, $\bar{b} \approx 1$) this gives $\lambda^\star \approx 0.67$, close to the corrective-to-distortive transition near $\lambda \approx 0.5$–0.6 in the simulated phase diagram (Figure 5). Combined with slow unaided learning as $\lambda \to 0$, this makes collective accuracy non-monotone in λ and gives the phase diagram an analytical anchor. The threshold is *local*: it predicts when small confidence-weighted mass gaps become self-amplifying around the symmetric indifference state. It does not classify terminal regimes by itself; the global regime map is obtained by simulation (Section 6.2), and an endogenous test of the threshold's predictive content — without any imposed early lead — is given in Appendix G.

Remark 2 (Relation to quantal response and learning equilibria)**.** *The logistic choice (4) is a quantal best response to the value contrast, so a consensus rest point of the community recursion — action frequencies that reproduce themselves under the choice map — is a quantal-response fixed point (McKelvey and Palfrey, 1995). The corrective and distortive regimes are stable rest points and $\lambda^\star$ marks a bifurcation of the anticipatory map; wrong consensus is a self-confirming non-optimal rest point, the process analogue of failures of naive learning to aggregate information (Golub and Jackson, 2010). Agents here are non-strategic: they do not reason about others' reasoning, so higher-order (level-k / cognitive-hierarchy) beliefs play no role. That strategic layer — beliefs about a representative agent, and their equilibria — is developed in the companion paper on legitimacy gating.*

5 Simulation design

Observables and regimes. We measure group regret $\mathcal{R}_T = \sum_{i,t \le T}(\mu^\star - \mu_{A_{i,t}})$, time to consensus, the polarisation index $\Pi_t = |m_{1,t}(1) - m_{2,t}(1)|$, correction lag after an early wrong lead, and, for the quotient study, the micro–meso discrepancy. Terminal outcomes are classified into four operational regimes using a 0.9 mass threshold: *efficient consensus* (every community places ≥ 0.9 on $a^\star$), *wrong consensus* (every community places ≥ 0.9 on the suboptimal arm), *polarisation* (communities lock on different arms), and *unresolved* otherwise.

Monte Carlo and ablations. Each configuration is replicated over independent reward and diffusion draws; reported regime probabilities are frequencies over replications. The design varies the reward gap, the anticipatory weight λ, the retrospective weight η, cohesion and permeability. Five decisive ablations isolate the mechanism: $\lambda = 0$ (no anticipatory channel), $\eta = 0$ (no retrospective channel), constant learning rates, no confidence weighting of transmission, and the alternative balance-of-evidence confidence map.

5.1 Identification strategy and uncertainty

Because the model is a fully specified stochastic data-generating process, the effect of each mechanism component is a well-defined causal quantity, and we estimate it as such rather than reading it off level differences.

Estimands. We target three objects: (i) regime probabilities, the population frequencies of each terminal regime at a configuration; (ii) means of group regret and correction lag; and (iii) *counterfactual component effects*. Writing do(c=0) for the intervention that disables component c (the anticipatory channel, the retrospective channel, the credibility weighting, the learning-rate modulation, or the decision-time confidence map), the effect of c on an outcome Y at operating point θ is

$$\tau_c(\theta) = \mathbb{E}[Y \mid \theta] - \mathbb{E}[Y \mid \theta,\ \mathrm{do}(c{=}0)],$$

the difference between the factual model and the otherwise-identical model with c switched off. This is a structural, model-internal counterfactual: the intervention changes exactly one switch and nothing else.

Identification by common random numbers. We estimate τ_c by simulating the full model and the ablation under *common random numbers*. Each replication pre-generates the same exogenous random environment — a common uniform reward draw and a common decision-noise draw for every agent and trial, indexed by the same seed — and each structural variant then realises the reward of its chosen arm by thresholding that shared uniform at the arm's mean. The exogenous environment is thus held fixed across variants while the endogenous action histories are free to differ. The replication-paired difference $D_r = Y_r(\text{full}) - Y_r(\mathrm{do}(c{=}0))$ then identifies τ_c with the exogenous environment held fixed — a within-replication, ceteris-paribus contrast that is the simulation analogue of a within-subject randomised intervention. There is no confounding because the only thing that differs between the paired runs is the structural switch; coupling also reduces the variance of $\widehat{\tau}_c$ relative to independent runs. The population estimand is the one the finite-N ensemble approaches as the dynamics concentrate on their mean-field limit (Proposition 3; cf. stochastic-approximation theory (Borkar, 2008)).

Uncertainty reporting. Regime probabilities are empirical frequencies over independent Monte Carlo replications; we report Wilson 95% intervals for binary regime probabilities and Monte Carlo standard errors with nonparametric percentile-bootstrap 95% intervals (resampling replications) for regret and correction lag. Counterfactual contrasts $\widehat{\tau}_c$ carry paired-bootstrap 95% intervals (resampling replication indices jointly). Every reported probability and mean is accompanied by its replication count R.

Verification. Before any result is produced, the implementation passes a ten-point test contract (row-stochasticity of W and B; bounded confidence; bounded values; the simultaneous-update identity; seed reproducibility; the no-social baseline; isolated communities $B = I$; full mixing; exactness of the quotient field; and the self-weighting convention), together with the drift–diffusion validation and an exact agreement check between the fast block engine and a dense-W reference engine. Appendix C lists the tests.

Reproducibility and scale. All randomness is seeded. Results below are reported at a *standard* compute scale ($N = 400$ agents per two-community society, 120–300 replications per configuration, 13×13 phase grids); the replication package also ships a *publication* scale

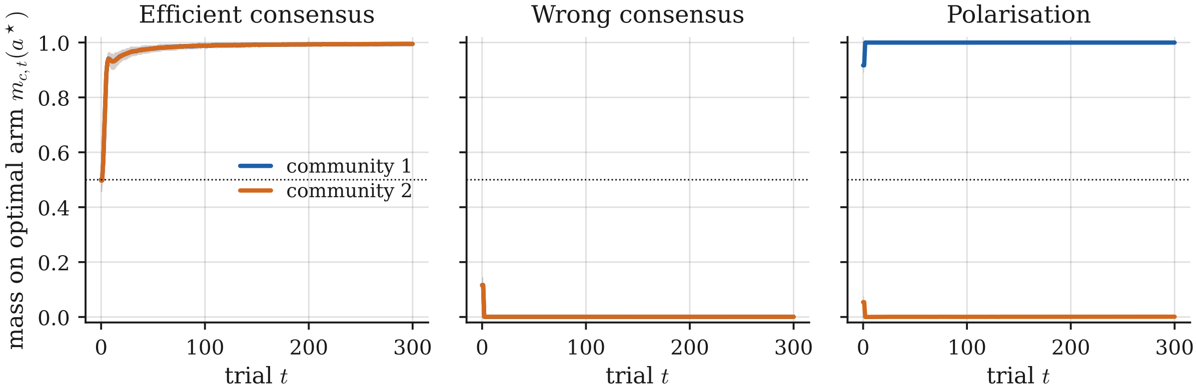

Figure 4: Baseline regimes. Mass on the optimal arm $m_{c,t}(a^\star)$ in each community (mean and 10–90 percentile band over $R = 300$ Monte Carlo replications; $N = 400$ agents, horizon $T = 300$). The same mechanism produces efficient consensus, wrong consensus and polarisation under the scenario parameters of Table 5.

($N = 1000$, up to 1000 replications, 21×21 grids) with identical mechanism parameters. The default parameters are listed in Appendix D.

6 Results

6.1 The mechanism produces four interpretable regimes

Figure 4 shows representative two-community trajectories of the mass placed on the optimal arm. With a clear reward gap and moderate transmission, both communities converge on the optimal arm (efficient consensus, $\mathbb{P} = 1.00$, mean regret 450). With a narrow gap, a strong anticipatory weight and an early confident lead on the suboptimal arm, the same machinery locks both communities onto the wrong arm (wrong consensus, $\mathbb{P} = 1.00$, regret **11,988**). With near-isolated communities and opposed early leads, the communities diverge and stay apart (polarisation, $\mathbb{P} = 1.00$, regret 5,998). Each regime occurs with probability 1.00 over $R = 300$ replications (Wilson 95% lower bound ≥ 0.987; Table 2). The ordering of regret across regimes — efficient < polarisation < wrong — already shows that the costliest outcome is not disagreement but confident, shared error.

6.2 Connectivity is non-monotone: the phase diagram

The central result is the connectivity phase diagram (Figure 5), which sweeps the anticipatory weight λ against cross-community permeability Γ in an asymmetric society where one community carries an early wrong lead. Three findings stand out. First, the effect of transmission strength is *non-monotone*. At $\lambda \approx 0$ lone learners fail to reach consensus within the horizon (unresolved); at moderate $\lambda \in [0.27, 0.53]$ transmission is *corrective* and produces efficient consensus across the whole permeability range; at large λ transmission becomes *distortive*. Second, the form of the distortion is governed by permeability. At high permeability the confident wrong community contaminates the correct one, yielding society-wide wrong consensus; at low permeability the communities insulate one another and the society polarises. Third, the distortive region dominates the high-λ plane: across the grid, wrong consensus is the modal outcome in 51% of the (λ, Γ) cells, efficient consensus in 24%, polarisation in 9% and unresolved in 15%. Connectivity is thus conditionally, not uniformly, beneficial.

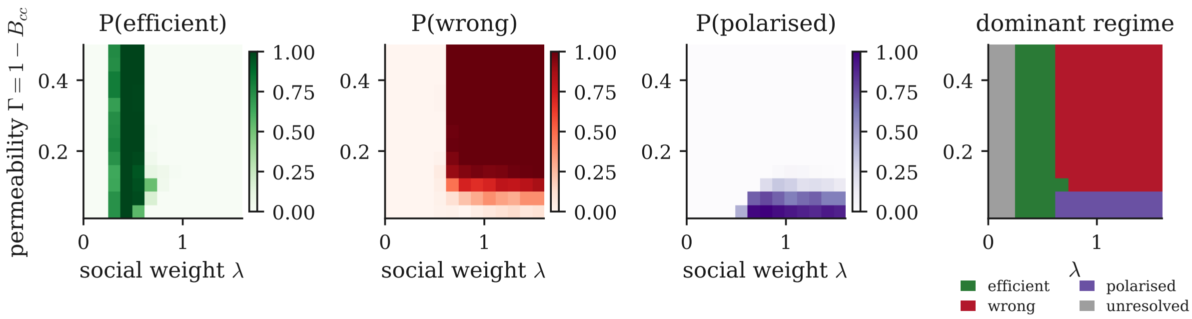

Figure 5: Connectivity phase diagram. Probability of efficient consensus, wrong consensus and polarisation, and the dominant regime, over the anticipatory weight λ and cross-community permeability $\Gamma = 1 - B_{cc}$. A corrective band at moderate λ separates an unresolved region ($\lambda \to 0$) from a distortive region (large λ); within the distortive region, low permeability yields polarisation and high permeability yields wrong consensus. Grid 13×13, 120 replications per cell, regime threshold 0.9, horizon $T = 240$; the first three panels show regime probabilities and the fourth the dominant (modal) regime. The Wilson half-width at 120 replications is at most 0.09.

6.3 A dynamical-systems view: bifurcation and phase portrait

The stochastic model admits an exact large-population ($N \to \infty$) reduction on balanced blocks (`credibility/meanfield.py`): within a community all agents share the same expected value table, choices are replaced by the exact DDM probability $\Lambda(2\beta a\Delta/\sigma^2)$, decision times by the mean first-passage time, and rewards by their means. Freezing values at the indifference point isolates the object of Proposition 3: the anticipatory feedback on the population lead $z_c = m_c(0) - \frac{1}{2}$. The symmetric mode has gain $\beta a\lambda\bar{b}\, C_0/\sigma^2$, which crosses one at $\lambda^\star = \sigma^2/(\beta a C_0 \bar{b})$; because the block coupling is row-stochastic, $\bar{b} = 1$ for a common lead and permeability enters only the anti-symmetric modes.

Figure 6 shows the resulting bifurcation: below $\lambda^\star$ the neutral state $z = 0$ is stable and small early leads decay (links are corrective); above $\lambda^\star$ it loses stability and a symmetric pair of amplified fixed points appears — $+z^\star$ a confidently correct consensus, $-z^\star$ a confidently wrong one — so the *same* links that corrected now lock in whichever early lead exists. The flow field (Figure 7) shows the same transition through its fixed points: a single attracting neutral state below $\lambda^\star$, and above it a repelling neutral state with four corner attractors (corrective, distortive, two polarised). The basin portrait (Figure 8) then shows this reorganisation of the flow across a sweep of λ. Below $\lambda^\star$ the entire early-lead plane drains to the neutral state: connectivity corrects any initial asymmetry. Above $\lambda^\star$ the neutral state becomes a saddle and the plane partitions into four basins of attraction — corrective, distortive and two polarised corners — so the *position* of the early lead, not its mere presence, now decides the collective outcome. As λ grows the corrective and distortive basins enlarge at the expense of the polarised ones. This is the mechanism of Figure 5 in miniature: the permeability Γ tilts the basin boundaries (a weakly coupled community resists an infected neighbour, favouring polarisation), while the finite-N fluctuations of the agent model scatter the early lead across the plane, so the basin areas become the regime frequencies measured there.

6.4 The wrong-consensus regime requires the full mechanism

We isolate each component using the counterfactual design of Section 5.1: the full model and five ablations are run under common random numbers ($R = 300$) at a contested operating point where

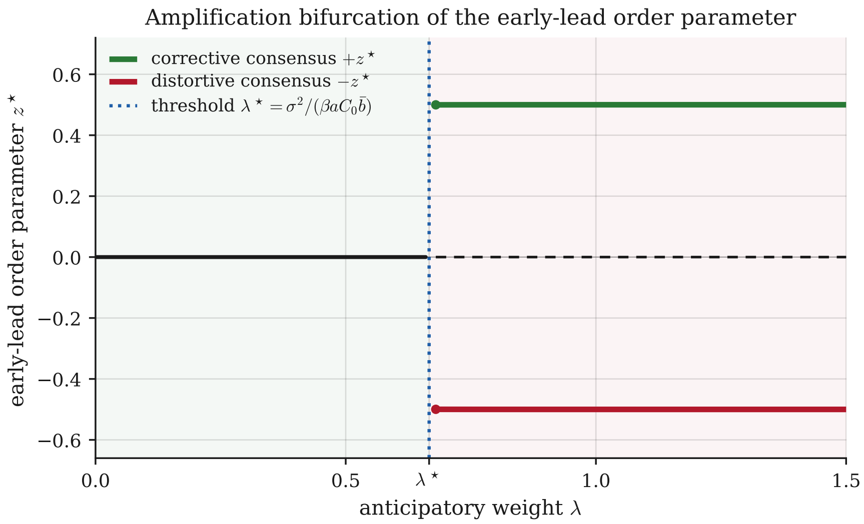

Figure 6: Amplification bifurcation of the mean-field reduction. The early-lead order parameter $z^\star$ is zero (stable neutral state) below $\lambda^\star$ and splits into a correct ($+z^\star$) and a wrong ($-z^\star$) amplified branch above it. The transition is mildly subcritical (a bistable window precedes $\lambda^\star$), so a sufficiently large early lead can lock in just below threshold. Small reward gap $|\Delta\mu| = 0.06$, $B_{cc} = 0.85$.

the full model produces wrong consensus with probability 0.83 (Wilson 95% interval $[0.78, 0.87]$). Figure 9 and Table 3 report the per-variant regimes and the paired counterfactual effects $\widehat{\tau}_c = \mathrm{P}(\text{wrong} \mid \text{full}) - \mathrm{P}(\text{wrong} \mid \text{ablation})$. The components fall into three classes, and every paired interval excludes zero. *Necessary* components, whose removal eliminates the distortion: disabling the anticipatory channel ($\lambda = 0$), the retrospective channel ($\eta = 0$), or the decision-time dependence of confidence (the balance-of-evidence map) each drives wrong consensus to 0.00, a paired effect $\widehat{\tau}_c = +0.83$ ($[0.79, 0.87]$) with group regret an order of magnitude lower. *Contributory but not necessary*: removing the credibility weighting of transmission *weakens but does not eliminate* the distortion (wrong consensus $0.83 \to 0.73$, $\widehat{\tau} = +0.10$, $[0.04, 0.16]$). *Buffering*: holding learning rates constant *intensifies* the distortion (wrong consensus $0.83 \to 0.99$, $\widehat{\tau} = -0.16$, $[-0.21, -0.12]$), because the confidence-modulated rate of (6), which enlarges corrections after confident mistakes, is the model's main private brake on error. The wrong-consensus regime is therefore not a generic feature of social learning: it is specific to the full confidence-weighted, dual-channel mechanism, and credibility weighting modulates rather than creates it.

6.5 Confidence attaches to error

Why can transmission distort? Because confidence is endogenous and can attach to wrong choices early in learning, exactly as the mechanism requires. Figure 10 shows the distribution of confidence conditional on whether the chosen arm was optimal, split by phase. Early in learning, wrong choices carry substantial confidence (mean 0.73, with appreciable mass above 0.9), only modestly below the confidence of correct choices (mean 0.98). These early confident errors are precisely the signals that the anticipatory channel amplifies before enough corrective evidence has accumulated. The mechanism requires no global irrationality — only that early stochastic success can generate confidence, and that confidence is socially read as credibility.

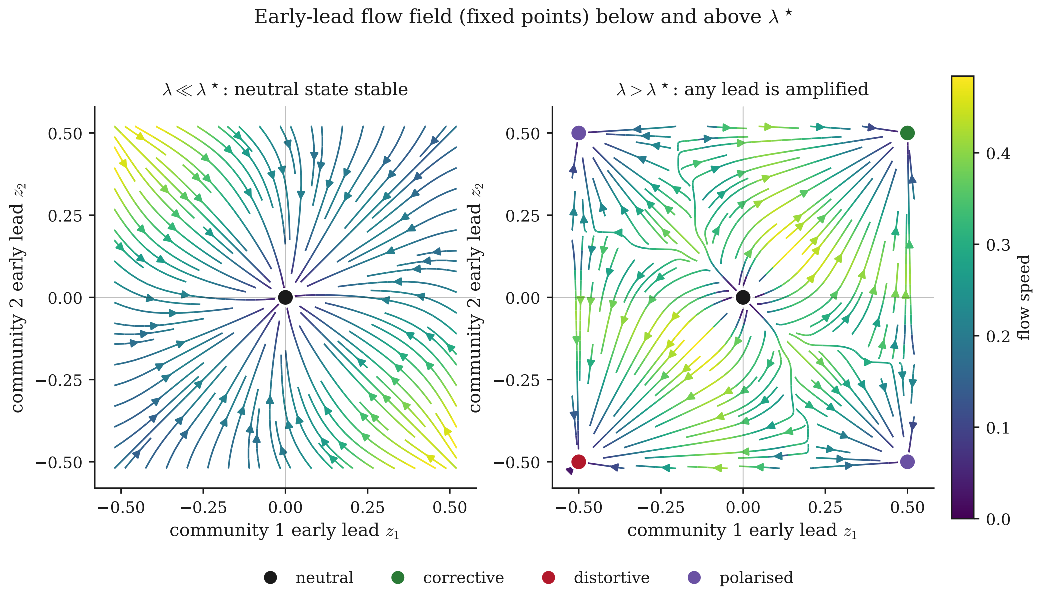

Figure 7: Early-lead flow field of the mean-field reduction, streamlines coloured by speed with the fixed points marked. **(left)** $\lambda \ll \lambda^\star$: the neutral state $(0, 0)$ is the sole attractor. **(right)** $\lambda > \lambda^\star$: the neutral state is a source and four attractors appear — corrective $(+, +)$, distortive $(-, -)$ and two polarised $(\pm, \mp)$ corners. Frozen-value reduction, $|\Delta\mu| = 0.06$, $B_{cc} = 0.85$.

6.6 The community quotient is faithful in the bulk and breaks at boundaries

Finally we validate the community-level (quotient) description against the full agent model (Figure 11). In two- and four-community examples the deterministic meso recursion tracks the micro ensemble mean closely, including a modular four-community society in which both descriptions reach the optimal arm (terminal micro mass 0.997 per community versus meso 1.000). Across a (λ, Γ) sweep the mean absolute discrepancy between the micro ensemble and the meso trajectory is 0.137, and the two descriptions agree on the dominant regime in 79% of cells. Consistent with Remark 1, the discrepancy is small in the interior of regimes and concentrates along the corrective/distortive phase boundary near $\lambda \approx 0.5$, where finite-N fluctuations decide the regime and the deterministic reduction is least reliable. The quotient is thus a useful but bounded instrument: exact for the anticipatory field, accurate for aggregate behaviour away from transitions, and explicitly not a substitute for the agent-level model near them.

6.7 Robustness

The phase structure is robust (Appendix F, Table 7, Figure 15). Across terminal thresholds (0.80/0.90/0.95), horizons (T up to 1000), reward gaps ($|\Delta\mu|$ from 0.02 to 0.20), and moderate perturbations away from balanced block weights (weighted stochastic-block and noisy row-normalised graphs), the corrective, distortive and polarised regimes persist, and the dominant-regime map is essentially unchanged across phase-grid resolutions and under a deterministic reaction-time approximation. The unresolved region at $\lambda \to 0$ is genuine rather than slow convergence: it persists at $T = 1000$. The principal sensitivity is to the strength of the early lead: at the distortive operating point even a small initial asymmetry ($\delta = 0.04$) is sufficient to produce wrong consensus, consistent with the amplification threshold of Proposition 3. The largest residual uncertainty sits, as expected, on the corrective/distortive boundary, where finite-N fluctuations and the quotient approximation are least reliable.

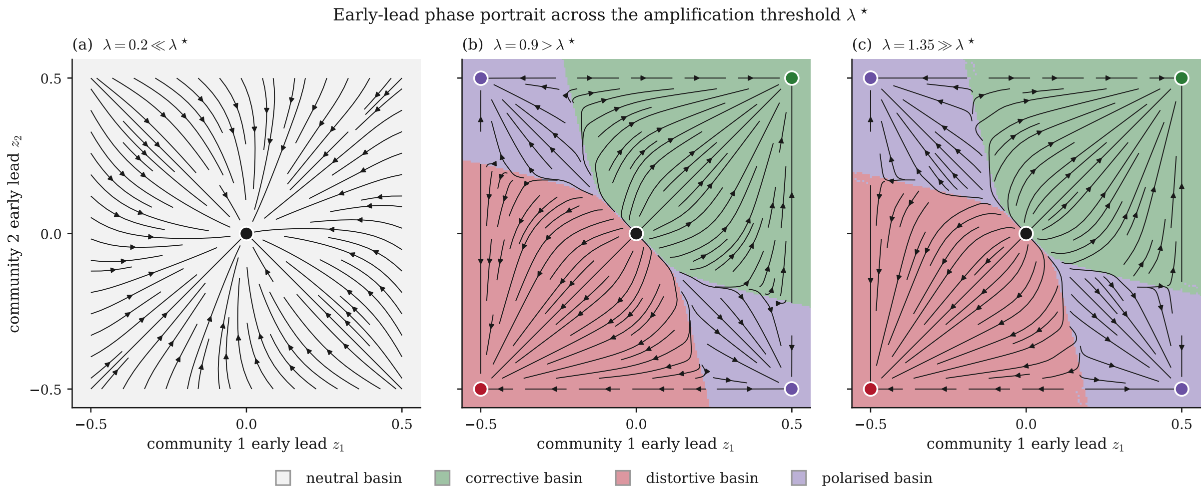

Figure 8: Early-lead phase portrait of the mean-field reduction across the amplification threshold. Axes are the two communities' early leads (z_1, z_2); streamlines show the flow and shading marks each point's basin of attraction (neutral, corrective, distortive, polarised). **(a)** $\lambda \ll \lambda^\star$: the whole plane drains to the neutral state — connectivity corrects any early asymmetry. **(b)** $\lambda > \lambda^\star$: the neutral state is a saddle and the plane splits into four basins, so the *location* of the early lead sets the outcome. **(c)** $\lambda \gg \lambda^\star$: the corrective and distortive basins expand and the polarised basins shrink. Frozen-value reduction, $|\Delta\mu| = 0.06$, $B_{cc} = 0.85$.

Table 2: Baseline regimes. Each scenario is run for R Monte Carlo replications; the realised regime probability is 1.00 in every case, reported with its Wilson 95% lower bound. $\mathcal{R}_T$ is mean group regret with Monte Carlo standard error. Scenario parameters are in Table 5.

Regime	R	λ	P(regime) [Wilson 95% LB]	$\mathcal{R}_T$ (SE)
efficient	300	0.40	1.00 [0.987]	450 (1)
wrong	300	0.90	1.00 [0.987]	11,988 (0)
polarised	300	0.90	1.00 [0.987]	5,998 (0)

7 Discussion

The simulations support four claims.

Connectivity is conditionally beneficial. The relationship between social transmission and collective accuracy is non-monotone (Figure 5). A corrective band of moderate credibility-weighted transmission separates an unresolved regime, in which agents have too little social information to converge, from a distortive regime, in which they have too much. The policy-relevant reading is that interventions which simply increase connectivity can move a society from the corrective band into the distortive region.

Confidence is an endogenous amplifier, not a report. Because confidence is produced by the decision process and can attach to error (Figure 10), social transmission is not neutral pooling. When credibility tracks accuracy it accelerates correction; when it tracks early, mistaken conviction it manufactures wrong consensus. The ablations (Figure 9) show that this depends on the specific way confidence is generated and used: the decision-time dependence of confidence and the confidence-modulated learning rate are not decorative.

Modularity both protects and entrenches. Low cross-community permeability prevents a confident wrong community from contaminating its neighbour, but the same insulation converts contagion into persistent polarisation (Figure 5). Modularity therefore trades one failure mode

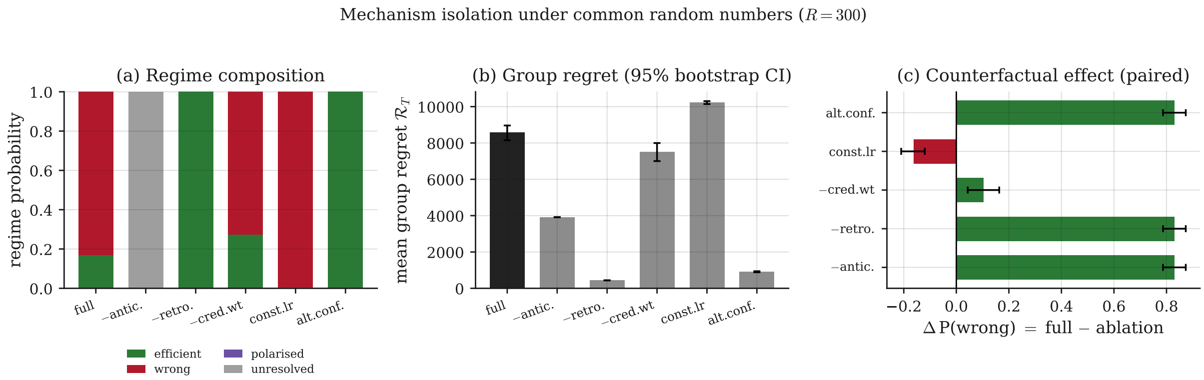

Figure 9: Mechanism isolation under common random numbers ($R = 300$). (a) Regime composition for the full model and five ablations at the contested operating point (Table 5); (b) mean group regret with 95% bootstrap intervals; (c) paired counterfactual effect on wrong consensus, ΔP(wrong) = full − ablation, with paired-bootstrap 95% intervals (green: the component is necessary for the distortion; red: the component buffers against it). Removing the anticipatory channel, the retrospective channel, or the decision-time confidence map eliminates wrong consensus; removing credibility weighting only weakens it; constant learning rates intensify it.

Table 3: Mechanism isolation at the contested operating point (Table 5), $R = 300$ replications under common random numbers. P(wrong) and P(efficient) carry Wilson 95% intervals; regret carries a 95% bootstrap interval. The last column is the paired counterfactual effect on wrong consensus, Δ =P(wrong|full)−P(wrong|ablation), with a paired bootstrap 95% interval. A positive Δ whose interval excludes 0 means the component is necessary for the distortion.

Variant	P(wrong) [95%]	P(eff.) [95%]	$\mathcal{R}_T$ [95%]	ΔP(wrong) [95%]
full	0.83 [0.78, 0.87]	0.17 [0.13, 0.22]	8,571 [8,148, 8,962]	—
no anticipatory	0.00 [0.00, 0.01]	0.00 [0.00, 0.01]	3,913 [3,908, 3,919]	+0.83 [+0.79, +0.87]
no retrospective	0.00 [0.00, 0.01]	1.00 [0.99, 1.00]	438 [433, 442]	+0.83 [+0.79, +0.87]
no credibility wt.	0.73 [0.67, 0.77]	0.27 [0.23, 0.33]	7,501 [7,002, 7,996]	+0.10 [+0.04, +0.16]
constant LR	0.99 [0.98, 1.00]	0.01 [0.00, 0.02]	10,239 [10,142, 10,303]	−0.16 [-0.21, -0.12]
alt. confidence	0.00 [0.00, 0.01]	1.00 [0.99, 1.00]	911 [882, 941]	+0.83 [+0.79, +0.87]

for another rather than removing failure.

Quotient aggregation is useful but limited. The community description is exact for the anticipatory social field (Proposition 2) and faithful for aggregate behaviour away from phase boundaries, but it is only an approximation of the stochastic value dynamics and degrades precisely where regime selection is decided (Figure 11). Mesoscopic reasoning about this class of models should be used for bulk behaviour and checked against the agent model near transitions.

Cognitive interpretation. The parameters carry direct cognitive readings: β is value sensitivity (how sharply value differences drive choice), σ internal decision noise, a the speed–accuracy threshold, κ_1, κ_2 the evidence- and time-dependence of confidence, α the reward learning rate, λ susceptibility to social influence, and η retrospective credit assignment (Appendix D). The retrospective channel is an other-referenced prediction error, the social counterpart of the reward prediction error (Joiner et al., 2017), and the confidence-modulated learning rule is a teaching-like signal in the sense of orbitofrontal adaptive-learning accounts (Schuck et al., 2016; Niv, 2019; Wang et al., 2023). The model tracks action values rather than latent task states; relating credibility transmission to belief-state or cognitive-map inference (Niv, 2019; Behrens et al.,

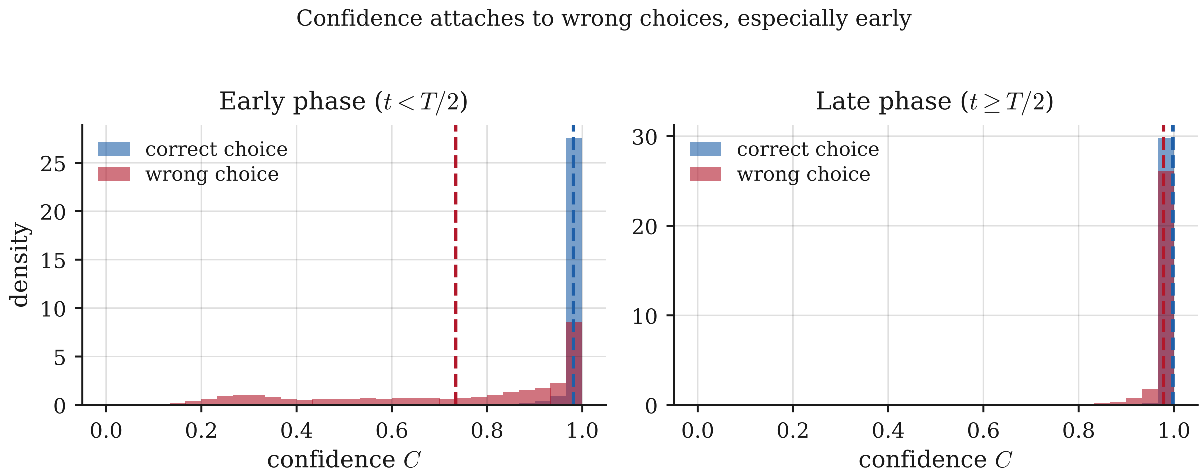

Figure 10: Endogenous confidence and early confident error. Distribution of confidence C for correct and wrong choices, early ($t < T/2$) and late ($t \geq T/2$), in a neutral contested scenario; densities are normalised to unit area and dashed lines mark group means. Early decisions number $\approx 1.8 \times 10^6$ (correct) and $\approx 6 \times 10^4$ (wrong). Early wrong choices carry a wide range of confidence, including high values (mean 0.73 versus 0.98 for correct choices): confidence attaches to error, which is what lets credibility-weighted transmission amplify it. The right panel shows that confidence concentrates near one late in learning as communities reach consensus.

2018) is a natural theoretical extension.

A normative benchmark. Group regret measures the summed reward shortfall relative to the omniscient optimum (zero regret). Against a *non-social* benchmark — the same agents with $\lambda = \eta = 0$ (pure reinforcement learning) at matched environments — credibility-weighted social learning is doubly signed: in the corrective regime it cuts mean regret from about 7,400 (non-social) to 450, a 16-fold improvement toward the ideal observer; in the distortive regime it inflates regret from about 4,800 (non-social) to 12,000, roughly 2.5 times worse. Confidence-weighted transmission thus approaches the optimum when credibility tracks accuracy and departs from both the optimum and the non-social learner when it amplifies early error — a precise statement of when the mechanism is beneficial versus pathological.

Testable predictions. The mechanism yields experiment-ready predictions. **(i)** High-confidence individuals should disproportionately influence collective learning, because credibility weights transmission by confidence. **(ii)** Early overconfident errors should produce persistent herd effects: above the local amplification threshold an early confident wrong lead locks in (Figure 16). **(iii)** Raising decision uncertainty — lowering confidence by degrading evidence or shortening deliberation — should reduce social contagion, since the anticipatory field is confidence-weighted. **(iv)** Manipulating confidence *independently of accuracy* (removing credibility weighting, or the balance-of-evidence confidence map) should change network dynamics without changing individual accuracy (Figure 9). **(v)** Because confidence is never observed directly but *inferred*, a receiver's weighting of a sender should track observable proxies of latent confidence — short reaction times, absence of choice vacillation, and past reliability — *controlling for* the sender's accuracy; masking those cues (hiding reaction time, adding response jitter, resetting reputation) should attenuate transmission even when accuracy is fixed. The confidence map itself makes this quantitative: since C falls with decision time (Eq. 5), following should decrease monotonically in a sender's reaction time at matched evidence. Each maps onto

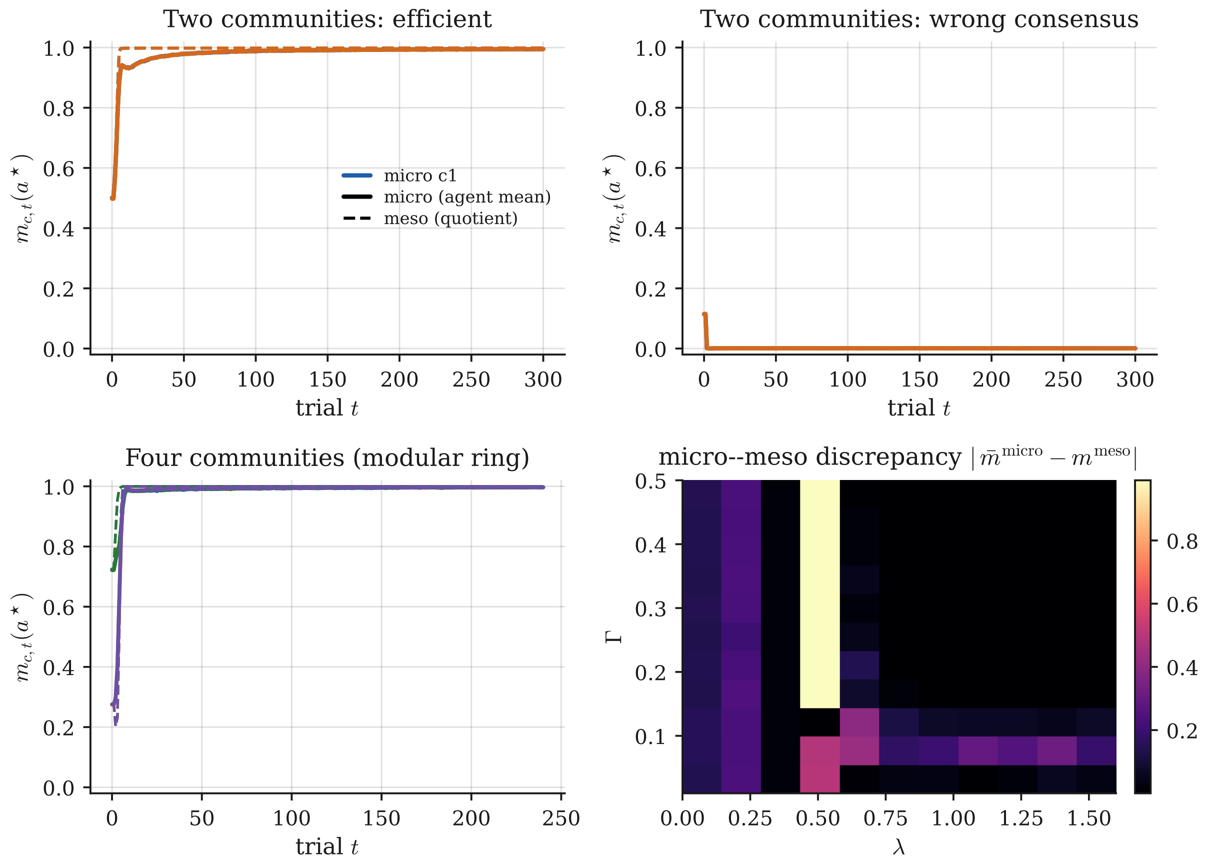

Figure 11: Micro–meso validation. Agent-level ensemble mean (solid) versus the deterministic quotient (dashed) for two-community efficient and wrong consensus and a four-community ($M = 4$) modular ring. Bottom-right: the micro–meso discrepancy $\frac{1}{M}\sum_c |\bar{m}^{\text{micro}}_{c,T}(a^\star) - m^{\text{meso}}_{c,T}(a^\star)|$ over an 11×11 (λ, Γ) grid (≈ 66 replications per cell). Mean discrepancy 0.137 and dominant-regime agreement 79%; the error concentrates along the corrective/distortive boundary near $\lambda \approx 0.5$.

a manipulable quantity in group-decision experiments.

A neuroeconomic reading. The model's variables line up, one for one, with quantities that decision neuroscience already measures (Figure 12): a value contrast driving an accumulator, a bounded-integration choice, a confidence readout, a reward prediction error, a confidence-gated learning gain, and a credibility-weighted *social* prediction error of the kind reported for other-referenced error signals (Joiner et al., 2017). The mapping is a functional analogy: the model commits to the computations, the substrates are candidate correlates. Its use is to turn predictions *(i)–(iv)* into a concrete, model-based experimental design (Figure 13). Confidence is manipulated orthogonally to accuracy — by speed pressure, evidence-quality framing, or an explicit confidence display — populating a confident-error cell that a purely accuracy-driven account cannot produce. The model then supplies the regressors (decision confidence $C_{i,t}$, private PE $\delta_{i,t}$, and the credibility-weighted social PE) for a model-based analysis of behaviour or neural signal, with the sharp prediction that a receiver's movement scales with sender confidence *controlling for* correctness. Because the transmitted confidence is a *latent* variable, the same design supports a hierarchical latent-variable analysis: estimate each sender's trial-by-trial confidence from reaction time, explicit report and choice history, then test whether the receiver's social prediction error is weighted by this *inferred* confidence rather than by observed accuracy.

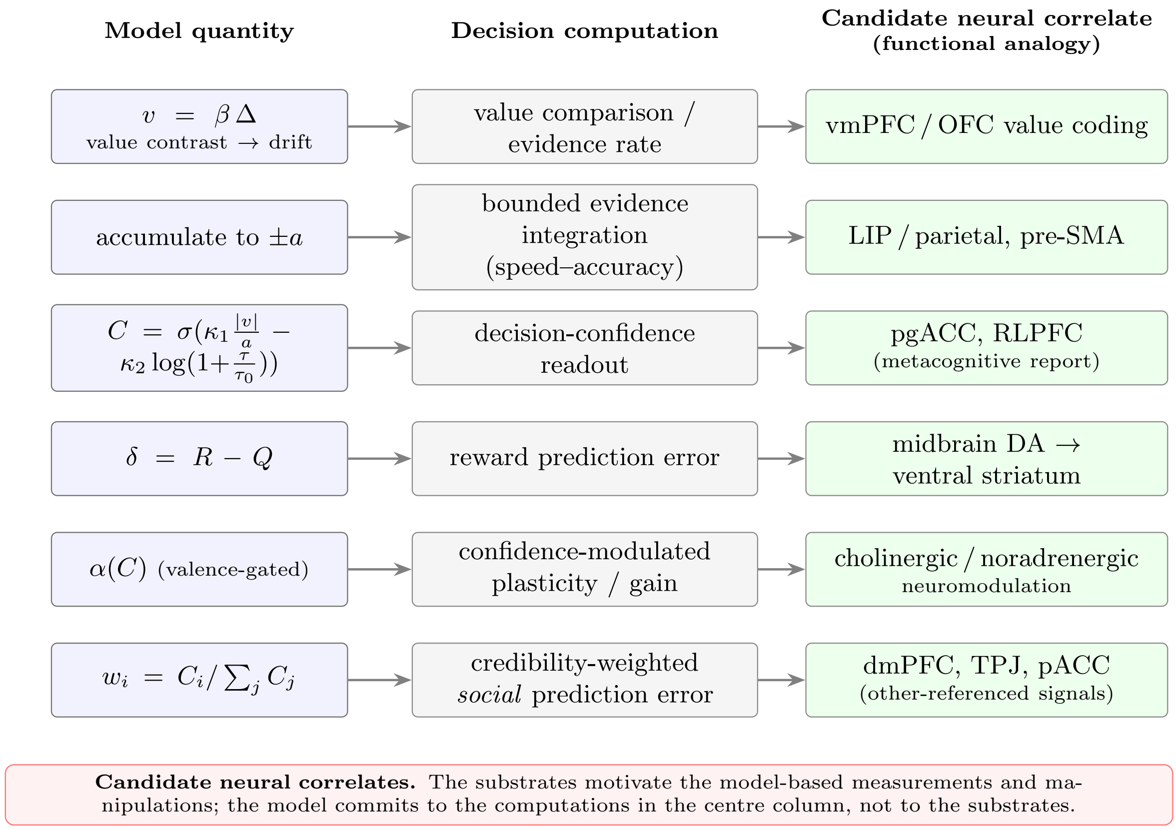

Figure 12: Computational-to-neural correspondence. Each model quantity (left) implements a decision computation (centre) with a candidate neural correlate (right). A functional analogy: it motivates the measurements and manipulations of Figure 13; the model commits to the centre column.

This turns the assumption that confidence is directly observed into an identifiable estimand and a falsifiable test of the credibility channel.

Limitations and scope. The baseline is deliberately narrow: binary actions, stationary rewards, fixed networks, community-homogeneous parameters, and a progression from two communities to a modular ring. Four assumptions are cognitively strong and each suggests an extension. Neighbours' confidence enters transmission as if directly observed, whereas confidence is usually *inferred* from reaction time, hesitation, or past reliability; we recast this as testable prediction (v) and a latent-variable estimand above, and a fully generative model that infers transmitted confidence from these cues is a natural next modelling step. The anticipatory and retrospective channels are kept independent, but exposure to a confident neighbour before a choice plausibly changes how later feedback is weighted, so the channels may interact. The network is fixed, whereas real influence weights adapt as agents learn whom to trust; endogenous, reputation-driven weights would make the model richer. And confidence here only modulates learning, whereas low confidence also promotes exploration and high confidence exploitation, a further behavioural role worth adding. Each extension should earn its place by the limitation it removes and the observable that identifies its effect, added only after the present mechanism is held fixed.

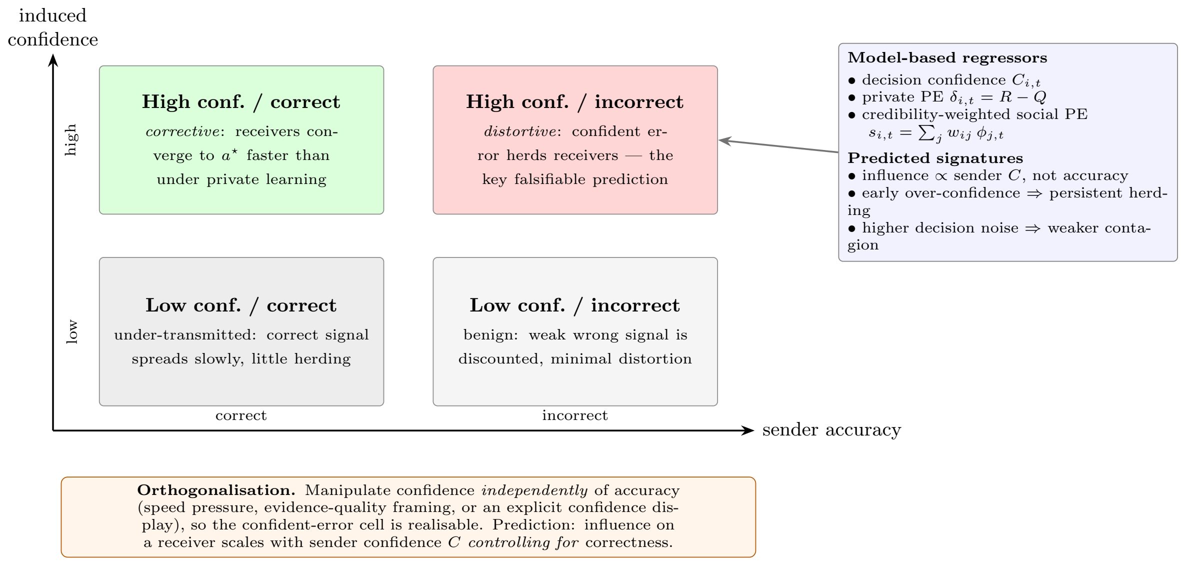

Figure 13: A model-based experimental design. Inducing confidence orthogonally to accuracy realises the confident-error cell (upper right), where the model predicts distortive transmission. The model supplies the regressors and falsifiable signatures (right), turning predictions *(i)*–*(iv)* into a testable protocol for dyadic or networked group-decision experiments.

Boundary of the present model. The model deliberately stops before legitimacy. Agents may be wrong, overconfident, or socially amplified, but they do not dispute the reward-generating rule itself. Realised rewards remain valid evidence for value revision. This boundary is useful because it isolates credibility transmission as a mechanism of collective error. It also identifies the next modelling step: if agents can discount rewards because they regard the rule as unfair, social learning becomes a joint process of value learning and legitimacy learning.

From credibility to legitimacy (future work). A natural extension is to relax the assumption that reward feedback is always treated as a legitimate learning signal. In the present model, prediction errors discipline latent values directly. In many social settings, however, a confident failure may be interpreted not as evidence that one's belief was wrong, but as evidence that the rule assigning rewards is unfair (Rabin, 1993; Fehr and Schmidt, 1999; Bolton and Ockenfels, 2000). This could be modelled by introducing a latent legitimacy state that gates the effective prediction error, together with a perceived fairness benchmark defined relative to a representative modal agent. Agents would then learn not only which actions pay, but also whether the payoff rule is acceptable. Network geometry would modulate the diffusion of both credibility and fairness benchmarks, potentially generating legitimacy polarisation, defensive wrong consensus, or correct but illegitimate consensus. This extension is left for future work.

8 Conclusion

We have presented an implemented agent-based model of social learning in which confidence is generated endogenously by a drift–diffusion decision process and transmitted socially as credibility through two distinct channels. The model is verified against an explicit test contract and studied by Monte Carlo simulation. Its central finding is that connectivity is non-monotone and conditional: moderate credibility-weighted transmission corrects error, while strong transmission amplifies early overconfident mistakes into wrong consensus or, under low permeability, into persistent polarisation. The same network links carry correction or distortion depending on whether the circulating credibility tracks accuracy or early error. A replication package regenerates every result. Future work can relax the assumption that rewards remain legitimate evidence and study when credibility-weighted communities learn not only which actions pay, but whether the payoff rule itself is acceptable.

Code and data availability

All code, configurations, fixed seeds and analysis scripts that reproduce every figure and table are released as the open-source CSL (Cognitive Social Learning) repository under an MIT license: https://github.com/<owner>/CSL (release tag and archive DOI to be inserted on publication). The repository documents the verification suite, the common-random-number ablation protocol, and one-command reproduction of the standard and publication profiles.

A Proofs

Proof of Proposition 1. Let $s(x) = e^{-2vx/\sigma^2}$ be the Wiener scale function; then $s(X_t)$ is a bounded local martingale up to the exit time τ of $(-a, a)$. Optional stopping gives $\mathbb{E}[s(X_\tau)] = s(0) = 1$, so with $p = \mathbb{P}(X_\tau = a)$, $p\,e^{-2va/\sigma^2} + (1-p)e^{2va/\sigma^2} = 1$, which solves to $p = 1/(1+e^{-2va/\sigma^2})$. For the mean time, $u(x) = \mathbb{E}_x[\tau]$ solves Dynkin's equation $\frac{\sigma^2}{2}u''(x)+v\,u'(x) = -1$ with $u(\pm a) = 0$. The general solution is $u(x) = -x/v + A + B\,e^{-2vx/\sigma^2}$; imposing the boundary conditions and evaluating at the origin yields $u(0) = A + B = (a/v)\tanh(av/\sigma^2)$. The limit $v \to 0$ follows from $\tanh(z)/z \to 1$. □

Proof of Proposition 2. By (1) and Assumption 1,

$$S_{i,t}(a) = \sum_d \sum_{j\in V_d} \frac{B_{cd}}{N_d} \mathbf{1}\{A_{j,t-1} = a\}\, C_{j,t-1} = \sum_d B_{cd}\Big(\frac{1}{N_d}\sum_{j\in V_d} \mathbf{1}\{A_{j,t-1} = a\}\, C_{j,t-1}\Big),$$

and the bracketed term is $\phi_{d,t-1}(a)$ by (9). The right-hand side depends on i only through its community c. □

Proof of Proposition 3. Near indifference the choice fraction $m_{c,t}(1) = \Lambda(2\beta a\bar{\Delta}_{c,t}/\sigma^2)$ has derivative $\Lambda'(0)\cdot 2\beta a/\sigma^2 = \beta a/(2\sigma^2)$, so the action-mass gap is $2m_{c,t}(1) - 1 = \kappa\,\bar{\Delta}_{c,t} + O(\bar{\Delta}_{c,t}^2)$. Both arms carry the indifference confidence C_0 to first order, giving $\psi_{c,t} = C_0\kappa\,\bar{\Delta}_{c,t}$. Substituting the mesoscopic contrast $\bar{\Delta}_{c,t} = \bar{\Delta}^Q_{c,t} + \lambda\bar{b}\,\psi_{c,t-1}$, which is the quotient field (10) written as a gap, yields $\psi_{c,t} = C_0\kappa\lambda\bar{b}\,\psi_{c,t-1} + C_0\kappa\,\bar{\Delta}^Q_{c,t}$. Holding the slowly varying value gap fixed, this is a scalar linear recursion with multiplier $\rho(\lambda) = C_0\kappa\bar{b}\lambda$, which decays geometrically iff $\rho < 1$. □

B ODD+D model description

A.1 Purpose and patterns. The model's purpose is to determine when credibility-weighted social transmission makes connectivity corrective and when it makes it distortive. It is designed to reproduce the patterns **P1** efficient consensus under moderate transmission; **P2** wrong consensus under strong transmission after an early confident error; **P3** polarisation under low cross-community permeability; **P4** a non-monotone effect of the social weight; **P5** quotient accuracy away from phase boundaries; and **P6** quotient degradation near phase boundaries. It is *not* designed to reproduce empirical opinion time series, continuous-action choice, or non-stationary environments.

A.2 Entities, state variables and scales. Entities are agents, communities, two arms and the weighted network. Time is discrete at the trial level with horizon T. Agent state: latent values $Q_{i,t}(a)$, social signal $S_{i,t}(a)$, augmented value $V_{i,t}(a)$, action $A_{i,t}$, decision time $\tau_{i,t}$, confidence $C_{i,t}$. Community state: action mass $m_{c,t}(a)$, confidence-weighted mass $\phi_{c,t}(a)$, mean value $\bar{Q}_{c,t}(a)$; the network is summarised by the row-stochastic coupling matrix B.

A.3 Process overview and scheduling. Each trial executes, synchronously across agents: (1) construct the anticipatory signal from the previous period; (2) form augmented values; (3) generate the drift–diffusion choice; (4) draw the decision time; (5) compute confidence; (6) draw rewards; (7) compute all private and social prediction errors from the *pre-update* values; (8) apply the private and social updates *simultaneously*; (9) record observables. The update is synchronous and simultaneous by construction (tested item 4, Appendix C).

A.4 Design concepts. *Basic principles*: reward-prediction-error learning, process-based choice, endogenous confidence, dual-channel sociality. *Emergence*: regimes and the phase structure emerge from agent rules and coupling, not from imposed aggregates. *Adaptation*: value revision via private and social prediction errors. *Objectives*: agents pursue reward myopically through learned values; there is no global optimisation. *Learning*: confidence-modulated private and social learning rates. *Prediction*: agents act on current augmented values; no forward simulation. *Sensing*: neighbours' previous actions and confidence (anticipatory) and current outcomes and confidence (retrospective). *Interaction*: weighted by W/B and by credibility. *Stochasticity*: diffusion noise and Bernoulli rewards. *Collectives*: communities. *Observation*: the observables of Section 5.

A.5 Initialisation. Initial values are community-constant; early leads are imposed through the initial value asymmetry. Community sizes, coupling matrix, reward means, horizon and seeds are listed per scenario in Table 5. Confidence is not initialised (it is generated at each trial).

A.6 Input data. None. The model is a synthetic computational mechanism model and uses no external empirical data.

A.7 Submodels. (1) anticipatory signal (1); (2) augmented value (2); (3) DDM choice (4) (Proposition 1); (4) reaction-time approximation (Section 3, Appendix C); (5) confidence (5); (6) private learning (6); (7) retrospective social learning (7); (8) projection to $[0, 1]$ (8); (9) regime classification (Section 5); (10) quotient approximation (10). Each maps to a code file and test in Table 6.

A.8 Parameters. Global default parameters are in Table 4; per-figure scenario parameters in Table 5.

A.9 Code-location map. Table 6 maps each model element to its equation, code file and test.

A.10 Verification, validation and reproducibility. See Appendix C for the test contract and drift–diffusion validation, and the replication metadata at the end of the main text for exact commands.

Table 4: Global default parameters (`code/credibility/config.py`). Confidence map is `decision` (Eq. 5); values are projected to $[0,1]$ each trial; reaction times use the inverse-Gaussian approximation.

Symbol	Default	Role	Config key
β	6.0	value sensitivity (drift scaling)	`beta`
σ	1.0	internal decision noise	`sigma`
a	1.0	speed–accuracy threshold	`a_thr`
κ_1	3.0	confidence: evidence weight	`kappa1`
κ_2	1.0	confidence: decision-time weight	`kappa2`
τ_0	0.5	reaction-time scale	`tau0`
$\alpha^{\min}$	0.05	min reward learning rate	`alpha_min`
$\alpha^{\max}$	0.4	max reward learning rate	`alpha_max`
γ	0.5	retrospective credit-assignment scale	`gamma`
ω	1.0	social learning-rate exponent	`omega`
ε	0.001	$\bar{C}$ denominator floor	`eps_soc`
RT disp.	0.3	reaction-time squared CV	`rt_dispersion`

Table 5: Scenario parameters for all reported figures (standard scale). N is total agents, T horizon, R replications, $\Delta\mu = \mu_1 - \mu_2$, λ anticipatory weight, η retrospective weight, $\Gamma = 1 - B_{cc}$ permeability. Regime threshold 0.9 throughout. Configuration: `code/credibility/scenarios.py`.

Scenario	N	T	R	$\Delta\mu$	λ	η	Γ	init. lead
Fig. 2 efficient	400	300	300	0.20	0.40	0.30	0.30	neutral
Fig. 2 wrong	400	300	300	0.10	0.90	0.20	0.30	asym.
Fig. 2 polarised	400	300	300	0.10	0.90	0.25	0.02	asym.
Fig. 3 phase (λ, Γ swept)	160	240	120	0.10	[0,1.6]	0.25	[.01,.5]	asym.
Fig. 4 ablation point	400	260	300	0.10	0.60	0.30	0.15	asym.
Fig. 5 confidence	300	260	60	0.10	0.50	0.25	0.15	neutral
Fig. 6 four-community	320	240	250	0.10	0.60	0.25	0.30	asym.

C Verification suite and drift–diffusion validation

The implementation passes the following automated tests (all green in the replication package): (1) row-stochasticity of W and B; (2) confidence bounded in $(0,1)$; (3) values bounded in $[0,1]$; (4) the simultaneous-update identity, checked against an independent reference and shown to differ from a sequential update; (5) seed reproducibility; (6) the no-social baseline, which is invariant to B when $\lambda = \eta = 0$ and learns the better arm; (7) decoupling of communities when $B = I$; (8) exchangeability under full mixing; (9) exactness of the quotient field (10) under balanced block weights, checked against explicit weighting; (10) the self-weighting convention (anticipatory includes self, retrospective excludes self). In addition, the exact choice probability (4) and the analytic mean decision time of Proposition 1 are validated against an Euler–Maruyama simulation of (3) (the empirical choice frequency matches (4) to within 0.025 and the empirical mean reaction time to within 8% across the drift range used), and the fast balanced-block engine reproduces the dense-W reference engine to machine precision on a balanced network. Appendix F (check R7) further shows that regime classification is insensitive to the reaction-time approximation.

Table 6: Equation-to-code-to-test map. Code paths are under `code/credibility/`; tests under `code/tests/test_credibility.py`.

Model element	Equation	Code	Test
Anticipatory signal	Eq. 1	`model.py: anticipatory_field_block`	`test_quotient_field_exact`
Augmented value	Eq. 2	`model.py: simulate_blocks`	`test_block_dense_*`
DDM choice	Eq. 4	`ddm.py: choice_prob_up`	`test_ddm_choice_*`
Reaction time	§3	`ddm.py: sample_decision_time`	`test_ddm_mean_fpt_*`
Confidence	Eq. 5	`confidence.py`	`test_confidence_bounded`
Private learning	Eq. 6	`model.py: _private_lr`	`test_simultaneous_*`
Retrospective learning	Eq. 7	`model.py: value_update_*`	`test_self_weighting_*`
Bounded values	Eq. 8	`model.py: np.clip`	`test_values_bounded`
Regime classification	§5	`metrics.py: classify_regime`	`test_regime_classification`
Quotient (meso)	Eq. 10	`model.py: simulate_meso`	`test_meso_*`
Uncertainty	§5.1	`uncertainty.py`	`--`

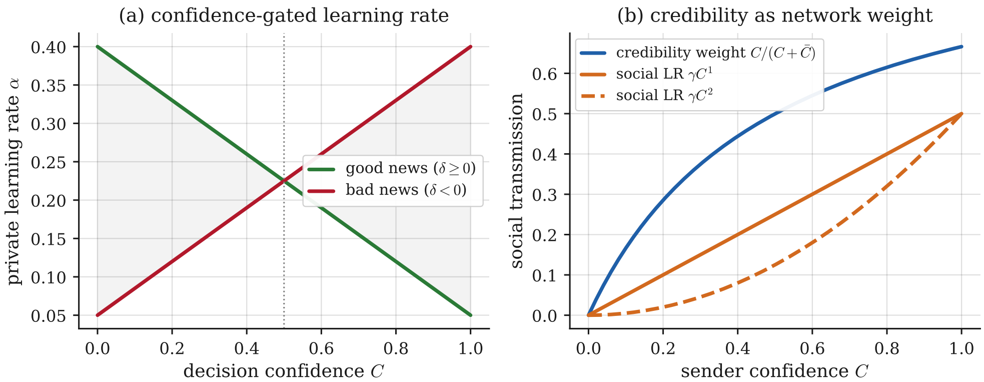

Figure 14: Quantitative companion to Figure 3. **(a)** the exact confidence-gated learning rates for good news ($\alpha_{\max} - (\alpha_{\max} - \alpha_{\min})C$) and bad news ($\alpha_{\min} + (\alpha_{\max} - \alpha_{\min})C$), crossing at $C = \frac{1}{2}$. **(b)** the credibility weight $C/(C + \bar{C})$ and the social learning rate γC^{ω} ($\omega = 1, 2$) versus sender confidence.

D Parameters, scenarios and code map

E Confidence-modulated learning and transmission

Figure 3 in the main text is a schematic; the exact relationships behind it are plotted here. The private learning rate (6) is valence-asymmetric in confidence (Figure 14a): the good-news and bad-news rates cross at $C = \frac{1}{2}$, so a low-confidence agent weights success and a high-confidence agent weights failure. The same confidence sets a sender's credibility weight $C/(C + \bar{C})$ and the social learning rate $\gamma\, C^{\omega}$ (Figure 14b) — the intra-agent and inter-agent roles of the one quantity.

F Robustness

We probe seven sensitivities (full CSVs in `paper/tables/`; Table 7, Figure 15). **R1** terminal threshold $\{0.80, 0.90, 0.95\}$: dominant-regime area fractions are invariant. **R2** horizon $T \in \{200, 300, 500, 1000\}$: the unresolved region at $\lambda \to 0$ persists (genuine, not slow convergence); corrective and distortive points are stable. **R3** early-lead strength δ: a small lead ($\delta = 0.04$) already triggers wrong consensus at the distortive point. **R4** reward gap $|\Delta\mu| \in \{0.02, \ldots, 0.20\}$:

Table 7: Robustness summary (Appendix F). Entries are the dominant-regime area fractions of the (λ, Γ) plane, or P(wrong) at the operating point, as indicated. Full tables: `paper/tables/robustness_*.csv`.

Check	Variation	Result
R1 threshold	0.80/0.90/0.95	wrong-area = 0.53/0.53/0.53 (invariant)
R2 horizon	$T = 200-1000$	near-zero stays unresolved; corrective/distortive stable
R3 initial lead	$\delta = 0-0.16$	P(wrong) = 0.12→1.00 (small lead suffices)
R4 reward gap	$\lvert\Delta\mu\rvert = 0.02-0.20$	P(wrong) = 1.00−1.00 (persists)
R5 random graphs	balanced/SBM/noisy	distortive P(wrong) = 1.00/1.00/1.00
R6 grid resolution	$7^2/9^2/13^2$	wrong-area = 0.49/0.53/0.51
R7 RT approximation	inverse-Gaussian vs. mean	wrong-area = 0.49 vs. 0.49

wrong consensus persists at the distortive point even for clearly distinguishable arms. **R5** random graphs: balanced block, weighted stochastic-block and noisy row-normalised W give the same regimes, so the results do not depend on exact quotient symmetry. **R6** phase-grid resolution $\{7^2, 9^2, 13^2\}$: the area fractions are stable. **R7** reaction-time approximation: the inverse-Gaussian and deterministic-mean reaction times give nearly identical regime maps.

G Endogenous early-lead test of the amplification threshold

Proposition 3 is a *local* statement: it predicts when small confidence-weighted mass gaps become self-amplifying, not the global regime map. We test its predictive content without any imposed lead. Agents start from neutral values ($Q = 0.5$ on both arms) with a small reward gap ($\mu = 0.53/0.47$) in two symmetric communities; a 25-trial burn-in lets a stochastic early asymmetry emerge, after which we measure the population confidence-weighted mass gap and the terminal regime ($R = 200$ per λ). Figure 16 shows that wrong consensus is *endogenous*: it is essentially absent for $\lambda \lesssim \lambda^\star$ and rises as λ crosses the analytical threshold $\lambda^\star \approx 0.67$, while lock-in fidelity is ≈ 1 — the terminal consensus arm is the burn-in lead arm, so the wrong-consensus replications are exactly those in which an early stochastic gap formed toward the suboptimal arm and was amplified. The threshold therefore predicts the onset of endogenous distortion, and the wrong-consensus regime does not depend on a hand-picked initial condition.

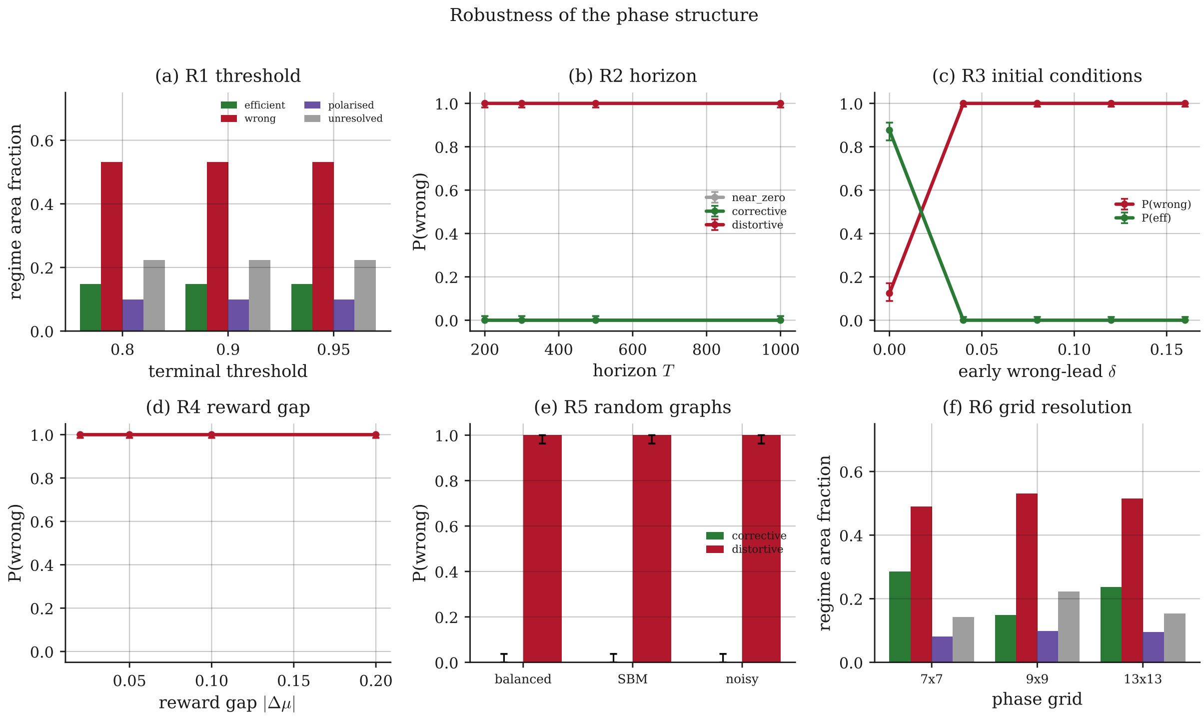

Figure 15: Robustness of the phase structure. (a) regime area fractions are invariant to the terminal threshold; (b) P(wrong) versus horizon for three operating points; (c) P(wrong) and P(efficient) versus early-lead strength δ; (d) P(wrong) versus reward gap with Wilson intervals; (e) P(wrong) under balanced, weighted-SBM and noisy graphs; (f) regime area fractions versus phase-grid resolution. Full tables in Table 7.

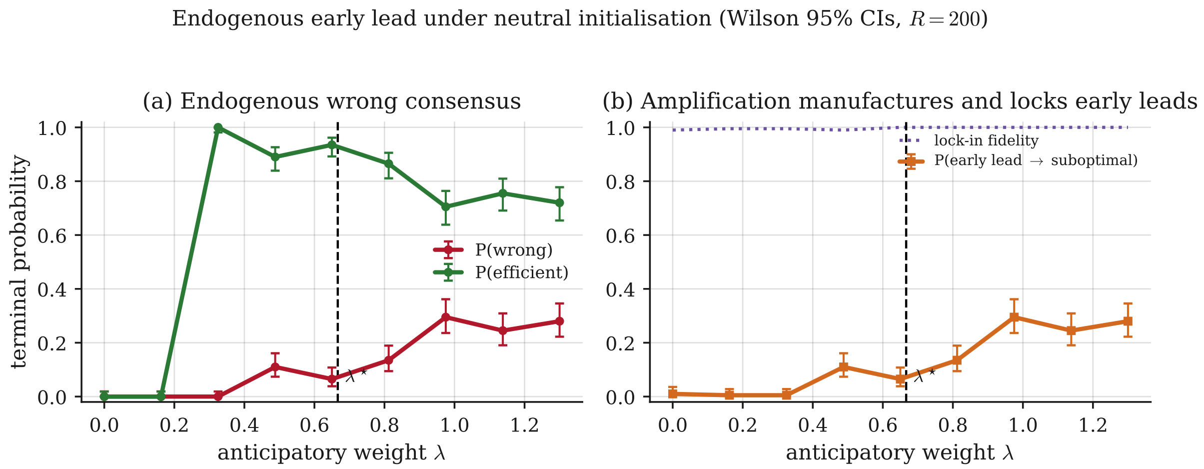

Figure 16: Endogenous early lead (neutral initialisation, $R = 200$ per λ). (a) P(wrong) is ≈ 0 below the local amplification threshold $\lambda^\star \approx 0.67$ and rises above it, mirrored by a fall in P(efficient). (b) the frequency of early suboptimal leads rises with λ and they are locked in with fidelity ≈ 1. Data: `tables/endogenous_early_lead.csv`.

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